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Device and method for continuous synthesis of graphene

Abstract

Provided herein is a method and a device for continuous synthesis of graphene. The device includes a container having a space for holding a carbon source, wherein the container has an entry opening for receiving the carbon source material, at least two electrodes for applying an electrical current through the space for joule heating the carbon source, wherein the space for joule heating the carbon source is between the at least two electrodes, and a movement component for moving the carbon source, with respect to the container, into the entry opening in a first direction and the at least two electrodes apply the electrical current in a second direction, wherein the first direction is not the same as the second direction.

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Background/Summary

TECHNICAL FIELD

(1) The embodiments disclosed herein relate to production of graphene, and, in particular to methods and devices for continuous production of graphene.

INTRODUCTION

(2) Graphene may be produced by conversion of a carbon source through joule heating. The carbon source is a carbon-based powder that is packed into a small quartz tube container and voltage is applied to each end of the powder material via metal (copper, copper wool, brass) electrodes. However, prior methods of synthesizing graphene produce several challenges in industrial applications and the method is not practical for mass production of graphene.

(3) Further, the quartz tubes that may be used for joule heating are cost-inefficient because they are degraded and contaminated during the joule heating process when the carbon powder comes into contact with the quartz. The quartz tubes must be disposed after a single use, which may significantly increase the cost of producing graphene on an industrial scale. The copper wool that is used as an electrode is also degraded by the process, which further increases the costs for producing graphene. Use of metal electrodes, such as copper and brass, which make direct contact with the graphene may also add metal contaminants into the graphene powder that is produced. Prior devices for synthesizing graphene also needed to be assembled and unassembled inbetween converting different batches of graphene which prevented large scale, continuous graphene production.

(4) Accordingly, there is a need for new cost efficient methods, products, and devices for producing graphene on an industrial scale. Methods, products, and devices of producing graphene which preserve the quartz used during synthesis may reduce costs. Further, products, methods, and devices are necessary to allow for continuous processing of materials necessary for producing graphene, thereby allowing for low-cost, industrial production of graphene.

SUMMARY

- (5) According to some embodiments, there is a device for synthesis of graphene. The device includes a container having a space for holding a carbon source. The container has an entry opening for receiving the carbon source material. The device includes at least two electrodes for applying an electrical current through the space for joule heating the carbon source. The space for joule heating the carbon source is between the at least two electrodes. The device also includes a movement component for moving the carbon source, with respect to the container, into the opening in a first direction and the at least two electrodes apply the electrical current in a second direction. The first direction is not the same as the second direction.
- (6) The device may also include an exit opening positioned relative to the at least two electrodes to allow for moving the graphene out of the space for joule heating the carbon source. The entry opening may be positioned relative to the at least two electrodes to allow for moving of the carbon source into the space for joule heating the carbon source while applying the electrical current.
- (7) The device may also include a power supply connected to the electrodes for passing the electrical current through the electrodes to convert at least part of the carbon source in the space for joule heating the carbon source into graphene.
- (8) The device may provide that the at least two electrodes are made from at least one of the group comprising copper, brass, stainless steel, and graphite.
- (9) The device may provide that the at least two electrodes have vents to allow for escape of gas when the electrical current is applied to the carbon source.
- (10) The device may provide that the space for joule heating the carbon source is surrounded by at least one quartz wall.
- (11) The device may provide that at least one of the at least two electrodes surrounds the space for joule heating the carbon source.
- (12) The device may provide that the at least two electrodes are resilient at high temperatures.
- (13) The device may provide that the at least two electrodes operate at temperatures between room temperature to 3200° C.
- (14) The device may also include pre-treatment electrodes for heating the carbon source to temperatures between 400° C.-800° C.
- (15) The device may provide that the at least two electrodes are configured to apply an electric current which heats the carbon source to temperatures between 2800° C.-3200° C.
- (16) The device may also include a carbon source reservoir for holding the carbon source prior to moving into the space for joule heating the carbon source.
- (17) The device may also include a graphene reservoir for collecting the graphene after moving out of the space for joule heating the carbon source.
- (18) The device may also include a compression component for compressing the carbon source.
- (19) The device may provide that compression component is a compression piston for compressing the carbon source.
- (20) The device may provide that the compression component is a compression corkscrew.
- (21) The device may provide that the electrode is at least one of the group comprising a ring electrode, a pin electrode, and a mesh electrode.
- (22) The device may provide that the at least two electrodes are positioned opposite to each other.
- (23) The device may provide that the at least two electrodes are positioned concentric to each other.
- (24) The device may provide that the electrical current is DC.
- (25) The device may provide that the electrical current is AC.
- (26) The device may provide that the electrical current is a combination of AC and DC.
- (27) Graphene may be produced by the device.
- (28) The graphene may be turbostratic graphene.
- (29) According to some embodiments, there is a device for synthesis of graphene. The device includes an inner electrode and an outer electrode. The device also includes a space for joule

heating the carbon source for a carbon source between the inner electrode and the outer electrode. The inner electrode and outer electrode are positioned to radially apply an electrical current through the space for joule heating the carbon source. The device also includes an entry opening positioned relative to the inner electrode and outer electrode to allow for moving of the carbon source into the space for joule heating the carbon source. The device also includes a power supply connected to the inner electrode and outer electrode for passing the electrical current through the inner electrode and outer electrode to convert at least part of the carbon source in the space for joule heating the carbon source into graphene.

(30) The device may provide that the outer electrode surrounds the space for joule heating the carbon source.

(31) The device may also include a cooling component for cooling at least one of the group comprising the inner electrode and the outer electrode.

(32) The device may provide that the electrical current is DC.

(33) The device may provide that the electrical current is AC.

(34) The device may provide that the electrical current is a combination of AC and DC.

(35) According to some embodiments, there is a device for synthesis of graphene. The device includes a container for holding a carbon source. The container includes a first electrode. The device also includes a space for joule heating the carbon source for applying an electrical current between a second electrode and the first electrode. The carbon source is positioned between the first electrode and second electrode. The device also includes a movement component for moving of the container to the space for joule heating the carbon source. The device also includes a power supply connected to the first electrode and the second electrode for passing the electrical current through the electrodes to convert at least part of the carbon source into graphene.

(36) The device may provide that the movement component is a conveyor belt.

(37) According to some embodiments there is a method for synthesizing graphene. The method includes moving a carbon source in a first direction into a space for joule heating the carbon source in a first direction. The method also includes applying an electrical current in a second direction to the carbon source with at least two electrodes positioned to allow movement of the carbon source into the space for joule heating the carbon source while applying the electrical current. The first direction and the second direction are not the same.

(38) The method may provide that applying an electric current to the carbon source includes applying a first electrical current to the carbon carbon source at a lower voltage for removing moisture and volatile materials from the carbon source. Applying an electric current to the carbon source may also include applying a second electrical current to the carbon source at a higher voltage for converting the carbon source to graphene.

(39) The method may provide that the lower voltage heats the carbon source to a temperature of between 400° C.-800° C.

(40) The method may provide that the electrical current is applied for between 50 milliseconds to about 1 second.

(41) The method may provide that the higher voltage heats the carbon source between 2800° C.-3200° C.

(42) The method may also include compressing the carbon source.

(43) The method may provide that the carbon source is compressed using a compression piston.

(44) The method may provide that the carbon source is compressed using a compression corkscrew.

(45) The method may also include removing unconverted carbon from the graphene.

(46) The method may provide that the carbon source is moved using a compression piston.

(47) The method may provide that the carbon source is moved using a conveyor belt.

(48) The method may provide that the at least two electrodes are positioned opposite to each other.

(49) The method may provide that the at least two electrodes are positioned concentric to each other.

- (50) The method may provide that the electrode is at least one of the group comprising a ring electrode, a pin electrode, and a mesh electrode.
- (51) The method may provide that the movement of the carbon source into the space for joule heating the carbon source is continuous.
- (52) The method may provide that the movement of the carbon source into the space for joule heating the carbon source is in batches.
- (53) The method may provide that the electrical current is DC.
- (54) The method may provide that the electrical current is AC.
- (55) The method may provide that the electrical current is a combination of AC and DC.
- (56) Graphene may be produced from the method.
- (57) The graphene of may be turbostratic graphene.
- (58) According to some embodiments, there is a device for synthesis of graphene. The device includes a container having a space for holding a carbon source, wherein the container has an entry opening for receiving the carbon source. The device also includes at least two ring electrodes for applying an electrical current through the space for joule heating the carbon source. The space for joule heating the carbon source is between the at least two ring electrodes. The device also includes a movement component for moving the carbon source, with respect to the container, into the entry opening in the same direction as the at least two ring electrodes apply the electrical current.
- (59) The device of may also include an exit opening positioned relative to the at least two ring electrodes to allow for moving the graphene out of the space for joule heating the carbon source. The entry opening may be positioned relative to the at least two electrodes while applying the electrical current to allow for moving of the carbon source into the space for joule heating the carbon source.
- (60) The device may also include a power supply connected to the electrodes for passing the electrical current through the at least two ring electrodes to convert at least part of the carbon source in the space for joule heating the carbon source into graphene.
- (61) The device may provide that the at least two electrodes are made from at least one of the group comprising copper, brass, stainless steel, and graphite.
- (62) The device may provide that the at least two electrodes include vents for escape of gas when the electrical current is applied to the carbon source.
- (63) The device may provide that the space for joule heating the carbon source is surrounded by at least one quartz wall.
- (64) The device may provide that the at least two ring electrodes are resilient at high temperatures.
- (65) The device may include pre-treatment ring electrodes for heating the carbon source to temperatures between 400° C.-800° C.
- (66) The device may provide that the at least two electrodes are configured to apply an electric current which heats the carbon source to temperatures between 2800° C.-3200° C.
- (67) The device may include a carbon source reservoir for holding the carbon source prior to moving into the space for joule heating the carbon source.
- (68) The device of may include a graphene reservoir for collecting the graphene after moving out of the space for joule heating the carbon source.
- (69) The device may provide that the movement component compresses the carbon source.
- (70) The device may provide that the movement component is a compression corkscrew.
- (71) The device may include a compression component for compressing the carbon source.
- (72) According to some embodiments, there is a method for synthesizing graphene. The method includes moving a carbon source in a first direction into a space for joule heating the carbon source. The method also includes applying an electrical current in second direction to the carbon source with at least two ring electrodes positioned to allow movement of the carbon source into the space for joule heating the carbon source. The first direction and the second direction are the same.
- (73) The method may provide that applying an electric current to the carbon source includes

applying a first electrical current to the carbon source at a lower voltage for removing moisture and volatile materials from the carbon source. The method may also include applying a second electrical current to the carbon source at a higher voltage for converting the carbon source to graphene.

(74) The method may provide that the lower voltage heats the carbon source to a temperature of between 400° C.-800° C.

(75) The method may provide that the electrical current is applied for between 50 milliseconds to about 1 second.

(76) The method may provide that the higher voltage heats the carbon source between 2800° C.-3200° C.

(77) The method may also include compressing the carbon source.

(78) The method may provide that the carbon source is compressed using a compression piston.

(79) The method may provide that the carbon source is compressed using a compression corkscrew.

(80) The method may also include removing unconverted carbon from the graphene.

(81) The method may provide that the carbon source is moved using a compression piston.

(82) The method may provide that the carbon source is moved using a conveyor belt.

(83) The method may provide that the at least two electrodes are positioned opposite to each other.

(84) The method may provide that the movement of the carbon source into the space for joule heating the carbon source is continuous.

(85) The method may provide that the movement of the carbon source into the space for joule heating the carbon source is in batches.

(86) The method may provide that the electrical current is DC.

(87) The method may provide that the electrical current is AC.

(88) The method may provide that the electrical current is a combination of AC and DC.

(89) Graphene may be produced from the method.

(90) The graphene may be turbostratic graphene.

(91) Other aspects and features will become apparent, to those ordinarily skilled in the art, upon review of the following description of some exemplary embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings included herewith are for illustrating various examples of articles, methods, and apparatuses of the present specification. In the drawings:

(2) FIG. 1 illustrates a diagram of a device for producing graphene in a laboratory setting, according to an embodiment;

(3) FIG. 2 illustrates a diagram indicating the directions of compression and electrode of a device for producing graphene in a laboratory setting, according to an embodiment;

(4) FIG. 3 illustrates a diagram indicating the directions of compression and electrode of a device with graphite disks for producing graphene in a laboratory setting, according to an embodiment;

(5) FIG. 4 illustrates a diagram of a device for continuous and batch synthesis of graphene, according to an embodiment;

(6) FIG. 5 illustrates a diagram of a device for continuous and batch graphene synthesis utilizing a piston-like action, according to an embodiment;

(7) FIG. 6 illustrates a diagram of a side view and top view of a device for continuous and batch graphene synthesis utilizing containers and electrodes from the top and bottom of the container, according to an embodiment;

(8) FIG. 7 illustrates a diagram of a device for continuous and batch synthesis of graphene utilizing a belt for graphene manufacturing through joule heating, according to an embodiment;

- (9) FIG. 8 illustrates a diagram of a side view of a device for continuous and batch synthesis of graphene utilizing corkscrew compression with split-ring electrodes being adjacent to the space for joule heating the carbon source, according to an embodiment;
- (10) FIG. 9 illustrates a diagram of a device for continuous and batch graphene synthesis utilizing corkscrew compression with mesh-like and ring-like electrodes adjacent the space for joule heating the carbon source, according to an embodiment;
- (11) FIG. 10 illustrates a device for continuous and batch synthesis of graphene utilizing ring-like electrodes adjacent to the space for joule heating the carbon source, according to an embodiment;
- (12) FIG. 11A is a diagram of a cross section view of a device for continuous and batch synthesis of graphene utilizing a radial electrode arrangement, according to an embodiment;
- (13) FIG. 11B is a diagram of a cross section view of a device for continuous and batch synthesis of graphene utilizing a radial electrode arrangement, and an extrusion screw which moves the carbon source in and out of the radial space of the device, according to an embodiment; and
- (14) FIG. 12 illustrates a flow chart of a method for synthesis of graphene, according to an embodiment.

DETAILED DESCRIPTION

- (15) Various apparatuses or processes will be described below to provide an example of each claimed embodiment. No embodiment described below limits any claimed embodiment and any claimed embodiment may cover processes or apparatuses that differ from those described below. The claimed embodiments are not limited to apparatuses or processes having all of the features of any one apparatus or process described below or to features common to multiple or all of the apparatuses described below.
- (16) Provided herein is a device and method for continuous synthesis of a graphene from a carbon source through joule heating.
- (17) The term “graphene” refers to a material which is a one-atom-thick planar sheet of sp²-bonded carbon atoms that are densely packed in a honeycomb crystal lattice, and, further, contains an intact ring structure of carbon atoms and aromatic bonds throughout at least a majority of the interior sheet and lacks significant oxidation modification of the carbon atoms. Graphene is distinguishable from graphene oxide in that it has a lower degree of oxygen containing groups such as OH, COOH and epoxide. The term “a graphene monolayer” refers to graphene that is a single layer of graphene. The term “a very few layer graphene” refers to a graphene that is between 1 to 3 layers of graphene. The term “a few layer graphene” refers to a graphene that is between 2 to 5 layers of graphene. The term “a multilayer graphene” refers to a graphene that is between 2 to 10 layers of graphene.
- (18) The term “turbostratic graphene” refers to a graphene that has little order between the graphene layers. Other terms which may be used include misoriented, twisted, rotated, rotationally faulted, and weakly coupled. The rotational stacking of turbostratic graphene helps mitigate interlayer coupling and increases interplanar spacing, thereby yielding superior physical properties relative to competitive graphene structures when compared on a similar weight basis. The subtle difference in adjacent layer stacking orientation expresses itself with important differences in product performance attributes. An important performance benefit evident with turbostratic graphene is that multi-layer graphene structures separate into few and individual graphene layers more easily and the graphene layers tend not to recouple. The turbostratic nature of a graphene may be observed and confirmed by Raman spectroscopy, Transmission Electron Microscopy (TEM), selected area electron diffraction (SAED), scanning transmission electron microscopy (STEM), and X-ray diffraction (XRD) analysis.
- (19) The term “opening” generally refers to any passage from which a carbon source or graphene may be passed through to enter or exit a space for joule heating the carbon source of a device for synthesizing graphene. The opening may include any an unobstructed entrance or exit. For example, an “entry opening” may be any entry point for moving a carbon source or graphene into a

space for joule heating the carbon source of a device for synthesizing graphene. An “exit opening” may be any exit point used for moving a carbon source or graphene out of a space for joule heating the carbon source of a device for synthesizing graphene.

(20) The term “carbon source” generally refers to any carbon-based material which may be converted into a graphene material, preferably turbostratic graphene. The carbon source may be in any form including in a powder form, or in a compressed pill form. The carbon source may include, without limitation, petroleum coke, tire carbon black, carbon black, metallurgical coke, plastic ash, plastic powder, ground coffee, anthracite coal, coal, corn starch, pine bark, polyethylene microwax, wax, chemplex 690, cellulose, naptenic oil, asphaltenes, gilsonite, and carbon nanotubes.

(21) In some embodiments, the devices, methods, and products provided herein may provide for continuous synthesis of graphene, wherein a carbon source is moved into a space for joule heating and converted to graphene and simultaneously moved out in a continuous manner. In some embodiments, the joule heating may be conducted in batches, wherein a first batch carbon source is moved into the joule heating space, the carbon source is converted to graphene, the carbon source is moved out of the joule heating space, and a new batch carbon source is moved into the joule heating space and the process is repeated. The joule heating in batches may be partial or full batches. In some embodiments, multiple partial batches may be moved into the joule heating space prior to moving out all of the batches. Similarly, multiple partial batches may be moved out of the joule heating space prior to moving in another batch.

(22) A flash joule heating synthesis method and compositions thereof are described in Patent Cooperation Treaty Application having International Publication Number WO 2020/051000 A1 to Tour et al., having an international publication date of Mar. 12, 2020, which is herein incorporated by reference in its entirety.

(23) A method of synthesizing graphene by joule heating a carbon pill and compositions thereof are described in Patent Cooperation Treaty Application having International Application Number PCT/CA2020/051368 to Mancevski, having an international application date of Oct. 13, 2020, which is herein incorporated by reference in its entirety.

(24) Referring to FIG. 1, illustrated therein is a diagram of a device for producing graphene in a laboratory setting, according to an embodiment. The device **100** for graphene synthesis consists of a mechanical assembly which includes a container **115** (for example a quartz tube) **115**, a the carbon source **105**, and electrodes **110** which may be made of copper wool or graphite. The mechanical assembly also includes compression pistons/screws and a protective enclosure which are not shown in FIG. 1. The device also includes an electrical assembly for providing an electrical current to the electrodes. The electrical assembly includes a power supply, a capacitor bank, high power switches, and a controller which are not shown in FIG. 1. The electrodes **110** make direct contact with the carbon source **105**. The carbon source **105** also makes contact with the quartz tube **115** wall, which causes the quartz tube **115** to degrade after the joule heating process. The joule heating current is up between 800 and 2500 Amperes. Due to they direct contact with the carbon source and the electrodes **110**, if the electrodes **110** are made from brass or copper, they may partially melt during the joule heating process and introduce imourities not the graphene.

(25) The device **100** of FIG. 1 may produce a batch of 1 g of graphene every ~15 min or 30 g/day from a variety of carbon sources. The quartz tube **115** has a 15 mm inner diameter (ID) with a carbon volume of ~5 cc. The joule heating process consists of loading the carbon powder into the quartz tube **115**. The process also includes plugging each of the quartz tube **115** ends with a layer of electrodes **110**. The process also includes compressing the carbon source with screws. Joule heating is performed for between 10 ms to 100 ms, and unloading the resulting graphene from the tube. In a laboratory setting this process is manual and takes about 15 min per batch. Most of the process time is overhead while the actual carbon-to-graphene process time is only 10 ms-150 ms. The energy required to convert carbon to graphene is 4 Wh/g (14.4 kJ/g) for fully discharging the joule heating process for 100 ms. Some of the energy is wasted because of the limiting speed of the high-

power electronics.

(26) Referring to FIG. 2, illustrated therein is a diagram indicating the directions of compression and electrode of a device for producing graphene in a laboratory setting, according to an embodiment. The direction of the electrodes **210** and the compression direction are predominately aligned. The carbon source **205** is compressed on both ends by the electrodes **210**. An advantage of the design shown in FIG. 2 is the possibility to use of standard round quartz tube **215**, the light emitted during joule heating (via radiation cooling process) of the carbon source **205** may exit through the quartz tube **215** to allow for reduced cooling times and gasses exit through the copper wool **210**. The device **200** results in very little loss of graphene material during handling. The device **200** also has disadvantages such as requiring disassembly to remove the graphene. Further, the graphene gets stuck to the copper and the copper wool particles end up mixed with graphene. In addition, the quartz tube **215** gets contaminated with each use.

(27) Referring to FIG. 3, illustrated therein is a diagram indicating the directions of compression and electrode of a device with graphite disks for producing graphene in a laboratory setting, according to an embodiment. To overcome the disadvantages of the device **200** of FIG. 2, a porous graphite plug or disk **220** is placed as the electrode component in direct contact to the carbon source **305**. Another plug made of copper wool **310** may optionally be added on each side of the graphite **320**, the graphite side that is not in contact with the carbon source. Then the copper wool **310** may be contacted by copper or brass rods that are connected to a DC power source such as a battery or capacitor or an AC power source. The carbon source **305** is compressed from one side in plane with the direction of the electrode. In some embodiments, the compression may be from both sides in plane with the direction of the electrode.

(28) The advantages of the device **300** is the possibility of using a standard round quartz tube **315**, the light emitted (radiation cooling) when joule heating the carbon source **305** may exit through quartz tube **315** to reduce cooling times and gasses may exit through the porous graphite and the copper wool. Furthermore, if the graphite is not porous, the process gasses may exit from the gap between the graphite plug and the inner diameter of the quartz tube. Further, the loss of material during handling is low using the device **300**. In addition, the graphene that is synthesized after the carbon source **305** is exposed to high temperatures of 2800° C. to 3200° C. and does not stick to the graphite plugs, or if the synthesized graphene does stick, the synthesized graphene may be easily removed from the graphite **320**.

(29) In some embodiments, the methods, devices, and products increase the batch size from 1 g to 10 g. Increase of the batch size directly increases the scale of the manufacturing process. The existing laboratory setup uses a quartz tube with a 15 mm ID and ~25 mm tube length, with a carbon mass of 1 g/batch, volume of ~5 cc/batch. Due to the resistive Joule heating nature of the load, doubling the tube diameter requires increasing the current by 4× with no changes in the voltage demand. Doubling the tube length requires an increase of the voltage by 2× with no changes in the current demand. Therefore, the batch size may be limited by the available switch electronics. Typical insulated-gate bipolar transistor (IGBT) high power switching modules, commonly used for motor drives, uninterruptible power supply (UPS) systems, and solar inverters have voltages of up to 1700V and currents up to 1800 A that allow for the flexible design of the reaction tube. Multiple IGBT switches, operating in parallel, may be used to enable currents up to 5000 A or more. The implication is that it is not practical to design a batch larger than 10 g. The scaling may be done by increasing the frequency (turn-around time) of each batch.

(30) In some embodiments, the methods, devices, and products provided herein achieve a batch size of 5 g per batch (volume of ~24 cc/batch) which equates to a production of 1 kg/day. This may be achieved with a quartz tube having 20 mm ID and ~75 mm tube length. The conversion power for 5 g is ~110 kW/g (1500 A and 750 V). Since the power is supplied by a bank of capacitors having a linear voltage discharge, the average voltage used for energy computations is ½ of the maximum voltage. In some embodiments, a batch size of 10 g per batch (volume of ~48 cc/batch)

may be possible, which allows for commercial production with the addition of automation and parallel production lines that will enable a production of 1 Ton/h (8 Ton/day). 10 g per batch may be achieved with a quartz tube having 20 mm ID and ~150 mm tube length. The conversion power for 10 g is ~110 kW/g (1500 A and 1500V).

(31) Referring to Table 1, a summary of device parameters for three different embodiments are shown with each allowing for larger scales of production. Embodiment 1 allows for the lowest scale of production of graphene. Embodiment 2 allows for larger scale production than Embodiment 1. Embodiment 3 allows for the largest scale production of the three embodiments and allows for industrial scale production.

(32) TABLE-US-00001 TABLE 1 Device Parameters for Scaling Graphene Production. Parameter Embodiment 1 Embodiment 2 Embodiment 3 Process Tube ID (cm) 1.5 2 2 Process Tube Area 1.77 3.14 3.14 (cm.sup.2) Process Tube Length 2.7 7.5 15.2 (cm) Process Volume (cc) ~5 ~24 ~48 Graphene Produced (g) 1 5 10 Process Current (A) 500 1500 1500 Process Voltage (V) 250 750 1500 Switch Module mech, relay IGBT IGBT Switch Specifications 900 V, 1000 V, 1600 V, 500 A, 25 ms 1600 A, 10 μs 1600 A, 10 μs

(33) In some embodiments, the methods, devices, and products provided herein decrease the process time (turn-around time) of each batch process for graphene synthesis. The decrease in process time relates directly with increases in the scalability of graphene production. With the devices of FIGS. 1, 2, and 3, the graphene synthesis process takes 15 min per batch of 1 g. Most of the process time is overhead while the actual carbon-to-graphene process time is only 10 ms-150 ms. The rest of the time is consumed with manual carbon loading, manual compression of the carbon powder, and manual graphene unloading. Using the devices in FIGS. 1, 2, and 3, 30 batches of 1 g of carbon source may produce 30 g of graphene per day.

(34) Production of 1 kg of graphene per day may be achieved with 200 batches of 5 g each. Assuming an 8 hour production day, production of 200 batches requires a budget of 2 min per batch. With a throughput of 2 min, the graphene synthesis process may be operated in a semi-batch mode where 10 or more batches are loaded at the same time and 5 g of carbon are undergo a joule-heating process in sequence without the need to load and unload.

(35) Similarly, production of 1 Ton/h may be achieved with 100,000 batches per hour, with each batch being 10 g each. Implementing 40 parallel production lines means that each production line needs to produce 20,000 batches per day, or 2500 batches of 10 g each per hour, assuming an 8 hour production day. Production of 2500 batches per hour requires a budget of 1.4 seconds per batch. The demand clearly requires a level of automation where the carbon is fed and removed into the process tube in a continuous manner while 10 g of carbon are joule-heated each 1.4 seconds without the need to load and unload. Alternatively, use of 100 parallel production lines is another reasonable possibility that increases the time budget to 3.6 seconds.

(36) Referring to Table 2, a summary of production parameters for the three embodiments of Table 1.

(37) TABLE-US-00002 TABLE 2 Production Parameters for Scaling Graphene Production. Parameter Embodiment 1 Embodiment 2 Embodiment 3 Batch Size (g) 1 5 10 No. Batches Per Day 30 20 800,000 Parallel Process Lines 1 1 40 No. Batches Per Line 30 200 20,000 Process Time Per 960 144 1.4 Batch (sec) Process Time Per 16.0 2.4 0.02 Batch (min)

(38) In some embodiments, the methods, devices, and products allow for a continuous process of graphene manufacturing. A continuous graphene synthesis method has the ability to load and unload the carbon powder in a continuous manner. A continuous graphene synthesis method also has the ability to compress the carbon powder independently from the loading and unloading. A continuous graphene synthesis method also has the orientation of the electrodes being independent from the direction of the carbon source flow. The continuous graphene synthesis method also has the availability of vent holes/pores so as to allow fast evacuation of hot gasses created as a result of the joule heating process. Also, there is a need to vent the reactor during the joule heating process

so that thermal and electrodynamic forces do not compress the carbon source.

(39) Referring to FIG. 4, illustrated therein is a diagram of a device for continuous synthesis of graphene, according to an embodiment. The device **400** allows for scaled production of graphene. The electrodes **410** are perpendicular to the carbon source **405** compression direction by the pistons **415**. As the carbon source **405** is pushed into the space for joule heating the carbon source **420**, which is right between the electrodes **410**, the carbon source may be joule heated to form graphene **425**. As new carbon powder is pushed into the entry opening **422** of the container **421** and into the space for joule heating the carbon source **420**, the resulting graphene **425** produced from the joule heating of the carbon source **405** may be expunged to the side opposite of where the new carbon source **405** is inserted. In some embodiments, there is no piston or other object on the exiting side of the tube as the carbon-tube friction will play the role of stopping the carbon from flowing and allowing joule heating to be commenced.

(40) An advantage of the device **400** of FIG. 4 is that the device enables continuous carbon source **405** and graphene **425** flow. The device **400** also does not need to be disassembled to insert the carbon source **405** or remove the graphene **425**. Further, the graphene does not stick to the graphite electrodes **410**. Low loss of graphene material occurs during handling.

(41) In some embodiments, the device for continuous synthesis of graphene includes a second set of electrodes for independent pre-treatment of the carbon source. The secondary set of electrodes, located left of the primary set of electrodes **410**, are used to pretreat the carbon at lower temperatures, between 400° C. to 1000° C. to carbonize the carbon source, remove moisture, volatiles, and any other substances that evaporate at these temperature ranges. After the carbon source is pre-treated the carbon source **405** is pushed towards the primary electrodes **410** where the carbon source **405** is joule heated and converted into graphene. The process tube may be any size, including without limitation round or rectangular. The process tube may be made of, without limitation, quartz or ceramic. The force needed to move large segments of graphene has been too large enough to move the carbon source.

(42) Referring to FIG. 5, illustrated therein is a diagram of a device for continuous graphene synthesis utilizing a piston-like action, according to an embodiment. The device **500** allows for scaled production of graphene. The device **500** includes a movement component **525** for moving the carbon source **505** into the entry opening **515** of the container **516**. The movement component is also for compressing the carbon source **505** and pushing the carbon source **505** towards the space for joule heating the carbon source **520**. The device **500** also includes a carbon source reservoir **535** for holding the carbon source **505** before the carbon source moves into the entry opening **515**. The electrical current is not in the same direction as the compression of the carbon source **505**. The electrical current is preferably perpendicular or close to perpendicular to the flow of carbon source **505**, however it may be in any direction which is not the same direction as the direction of compression. The flow of carbon **505** may be continuous and the electrodes **510** do not prevent the flow of the carbon source **505**. After the carbon source **505** is joule heated, the carbon source **505** is converted to graphene powder **506** that follows the flow path of the carbon source **505**. The converted graphene **506** moves out of the exit opening **545** and into a graphene reservoir **540**. Alternatively, pistons **525** and **530** may be replaced by a corkscrew or extrusion screws that rotate continuously instead of moving in forward-backward strokes.

(43) The device **500** of FIG. 5, may be constructed with oval or rectangular cross section quartz or ceramic tubes (~150×150 mm), in which there are cutouts for the parallel plate electrodes (~150×18 mm) in the mid-section of the tube. Alternatively, the electrodes may be curved to match the radius of curvature of the inlet and outlet conduits. The volume between the electrodes needs to fit 24 cc (5 g) or 48 cc (10 g) in accordance to the dimensions from Table 1. Unlike the device for producing graphene in a laboratory setting from FIG. 3, the function of the compression pistons **525**, **530** and the electrodes **510** in FIG. 5 is carried out by separate components. The electrodes **510** may be made from, without limitation, copper, stainless steel, graphite or tungsten. The electrodes **510** are

attached to the quartz but have vent holes to enable escape of the hot process gasses during the joule heating process. The compression pistons **525**, **530** may be made of a dielectric material, such as quartz or ceramic, to prevent shorting to the piston's ground. The carbon source **505** is fed through a carbon source reservoir **535** into the entry opening **515** of the process tube. The movement of the carbon source is aided by a shaker while the first compression piston is retracted to the left. After the carbon powder is dispensed in the tube, the first compression piston **525** moves with enough stroke to displace the converted carbon **506** from the space for joule heating the carbon source underneath the electrodes while at the same time the second compression piston retracts to the right to allow for the graphene to be emptied into a collection bin. The movement is aided by a shaker or a vacuum suction. After the stroke is over, the second compression piston **530** pushes in to block the tube while the first compression piston **525** applies a predetermined pressure to the carbon source **505** until the joule heating process is done. The piston cycles need to match the throughput of the continuous process. The flow of carbon source **505** and graphene material **506** is in an enclosed environment that makes the operation safe.

(44) Referring to FIG. **6**, illustrated therein is a diagram of a side view and top view of a device for continuous graphene synthesis utilizing containers and electrodes from the top and bottom of the containers **620**, according to an embodiment. The device **600** allows for scaled production of graphene. In this embodiment, the container **620** is not a tube. Instead the carbon source **605** is loaded into containers **620** through the entry opening and the electrodes **610** are the top and bottom of the container **620**.

(45) Advantages of the device of FIG. **6** include that the device **600** enables continuous carbon source **605**/graphene **606** flow. The device **600** does not need to be disassembled to insert the carbon source **605** or remove the graphene **606**. The graphene **606** does not get stuck to the graphite electrodes **610** and the device **600** allows use of second set of electrodes **610** for independent pre-treatment of the carbon source **605**. The secondary set of electrodes, located left of the primary set of electrodes **610** are used to pretreat the carbon source at lower temperatures between 400° C. to 1000° C. to carbonize the carbon source **605** and remove moisture, volatiles, and any other substances that evaporate at the pre-treatment temperature ranges. After the carbon source **605** is pre-treated the carbon source container **620** is moved under the primary electrodes **610** where the carbon source **605** is converted into graphene **606**.

(46) Referring to FIG. **7**, illustrated therein is a diagram of a device for continuous synthesis of graphene utilizing a belt for graphene manufacturing through joule heating, according to an embodiment. The device **700** allows for scaled production of graphene. The carbon source **705** is fed through a reservoir **725** into the entry opening **731** of the quartz or ceramic container **730**, having a metallic bottom electrode **711** that is grounded electrically, and is part of a continuous belt **735**. The quartz container **730** may be a cylinder with diameter of 20 mm and depth of 150 mm, with a pitch of 50 mm between boats. The piston/electrode **710** is a cylinder with diameter of >20 mm to cover and seal the top surface of the boat. The piston/electrode **710** may be copper, steel, tungsten, or graphite. Vent holes are included in the electrode **710**. The container **730** volume is 24 cc (5 g) or 48 cc (10 g) in accordance to the dimensions from Table 1. The container **730** is moved into the joule heating space **720** in a direction that is different than the direction the electrical current is applied by the electrodes **710**, **711**.

(47) The dispersion of the carbon source **705** is loose until the container **730** reaches the space under the piston/electrodes **710** that compresses the carbon source **705** and applies joule heating to the carbon source **705** to convert the carbon source **705** in the container **730** into graphene **706**. As the belt moves away from the space for joule heating the carbon source **720** the container **730** is emptied into a collection bin (not shown). The piston cycle is 2 minutes or 1.4 seconds to match the throughput of the continuous synthesis of graphene. The belt speed moves the next container under the electrode every 2 min or 1.4 s (depending on the desired scale). The entire device is enclosed to make the operation safe. The limiting step in the devices of FIGS. **6** and **7** is the speed of filling the

container **730** with carbon source **705** from the reservoir **725**.

(48) Referring to FIG. **8**, illustrated therein is a diagram of a side view of a device for continuous synthesis of graphene utilizing corkscrew or extrusion screw compression with split-ring electrodes being adjacent to the space for joule heating the carbon source, according to an embodiment. The device **800** allows for scaled production of graphene **806**. The carbon source **805** is moved into the entry opening **830** of the container **831** which is the extrusion component and compressed by corkscrew compression **815**. The corkscrew compression **815** pushes the carbon source **805** past two sets of electrodes **810**, **811** positioned on the side of the flow tube where the pre-treatment space **821** and space for joule heating the carbon source **820** are. The electrodes are positioned perpendicular to the flow of carbon source. The corkscrew **815** may be a single corkscrew or a twin corkscrew mechanism. The electrodes **810**, **811** are parallel plate **830** or opposing half-side electrodes **830** located opposite of each other at the same level of the extrusion component. A first set of electrodes **811** is used for pre-flashing the carbon source **805** and another set **810** for joule heating the carbon source **805**. The pre-treated carbon source moves in the direction of the space for joule heating the carbon source **820** where the second set of electrodes **810** are positioned. An optional spring-loaded lid **825** may hold the graphene powder **806** until the graphene powder **806** is ready to be removed. The opening of the lid **825** is synchronized with the motion of the corkscrew **815**. The electrical current is perpendicular to the flow of carbon source **805**.

(49) Alternative sets of electrodes **812**, **813** may be used for the device **800** of FIG. **8**. The ring electrode **813** may be used for one or more of the terminals with a pin electrode **812** used for at least one terminal. The cross section shows that the pre-treatment electrodes **811** at the pre-treatment space **821** are parallel opposing electrodes **830** and the joule heating electrodes **812**, **813** are ring electrode **813** for the negative terminal and the pin electrode **812** for the positive terminal. The ring **813** and pin **812** electrodes are concentric and will joule heat the carbon source **805** in the space for joule heating the carbon source **820** between the pin and the ring.

(50) Referring to FIG. **9**, illustrated therein is a diagram of a device for continuous graphene synthesis utilizing corkscrew compression with mesh-like and ring-like electrodes adjacent the space for joule heating the carbon source, according to an embodiment. The device **900** allows for scaled production of graphene **906**. The carbon source **905** is inserted into a corkscrew mechanism **915** of an extruder-like machine that compresses the carbon source **905** and pushes the carbon source **905** past two sets of electrodes **911**, **910** positioned upstream and downstream of the flow of carbon. The first set of electrodes **911** are mesh electrodes that allow flow of material but may also be energized and allow joule heating. The top mesh **930** is the ground electrode and the bottom electrode **931** is the positive electrode. The electrodes **930**, **931** may also be positioned in the opposite orientation where the positive electrode **931** is the top mesh and the ground electrode **930** is the bottom electrode. The first set of electrodes **911** are for pretreating the carbon source **905** at the pretreating space **921**. The second set of electrodes **910** are for joule heating the carbon source **905** at the space for joule heating the carbon source **920**. An optional spring-loaded lid **925** may hold the graphene powder **906** until the graphene powder **806** is ready to be removed. The opening of the lid **925** is synchronized with the motion of the corkscrew **915**. The electrical current is perpendicular to the flow of carbon source **905**. The carbon source **905** is moved into the entry opening **922** of the container **923**, which is the extrusion component.

(51) In another embodiment, shown in FIG. **9**, the electrodes **911**, **912**, **913**, **914** are ring electrodes where the ground electrode **913** is the top electrode and the bottom electrode **912** is the positive terminal. The electrodes **911**, **912**, **913**, **914** may also be positioned in the opposite orientation. The ring electrodes **911**, **912**, **913**, **914** may be half ring, semi-pole rings, or quadrature-pole electrodes for more uniform distribution of the current flow through the sample. An optional spring-loaded lid **925** may hold the graphene powder **906** until the graphene powder **906** is ready to be removed. The opening of the lid **925** has to be synchronized with the motion of the corkscrew **915**.

(52) In both examples shown in FIG. **9**, the carbon source **905** flows from top of the tube to the

bottom of the tube and the electrical current flows in the direction of the carbon flow but has an open flow design so as not to interfere with the carbon powder flow.

(53) Referring to FIG. 10, illustrated therein is a device for continuous synthesis of graphene utilizing ring-like electrodes adjacent to the space for joule heating the carbon source, according to an embodiment. The device **1000** allows for scaled production of graphene **1006**. The carbon source **1005** is moved using an extruder like machine and the electrodes **1025**, **1030** are ring electrodes located along the length of the compressed carbon source **1005** flow. One ring electrode **1030** is located right at the exit opening **1035** of the extruder and the other electrode **1025** is located further along the tube **1040** upstream of the carbon source **1005** flow adjacent the entry opening. In this embodiment, the tube **1040** is the container and the carbon source is moved into the entry opening **1026** of the tube in the same direction as electrical current from the ring electrodes **1025**, **1030**. The separation of the rings **1025**, **1030** determines the size of the space for joule heating the carbon source **1045**. The electrical current flows from the inside ring of the first electrode **1025** to the inside ring of the second electrode **1030**. During operation, the space for joule heating the carbon source **1045** between the electrodes is filled with carbon source **1005** and the joule heating of the carbon source **1005** is executed. The synthesized graphene **1006** in the space for joule heating the carbon source **1040** between the electrodes is removed through an exit opening **1035**. A restricting orifice **1050** is added to the exit opening **1035** to prevent unwanted flow of graphene powder **1006**. Additionally, the friction of the graphene powder **1006** in the tube **1040** prevents unwanted flow of graphene powder **1006**. The flow of carbon source **1005** may be in full batches where the space for joule heating the carbon source between the electrodes **1045** is completely filled. The flow of carbon source **1005** may also be in semi batches, where only a fraction of the graphene **1006** is removed and the same fraction of carbon source **1005** is filled back. The flow of carbon source **1005** may also be continuous where the electrical discharge is periodic with respect to a continuous motion of the carbon source **1005** and graphene **1006**.

(54) Referring to FIG. 11A, illustrated therein is a diagram of a cross section perspective view of a device for continuous synthesis of graphene utilizing a radial electrode arrangement, according to an embodiment. The device includes an inner electrode **1110** and outer electrode **1120**, that are preferably made of graphite. The electrical potential of this device is in radial direction, originating from the outer surface of the inner electrode **1110** and terminating to the inner surface of the outer electrode **1120**, or vice versa. The device **1100** includes an outer graphite electrode **1120** that acts as an electrode but also acts as a confinement tube to replace a quartz tube and may be cooled from the outside using conduction or convection cooling. The device may include a copper pipe **1105** having circulating coolant made from dielectric oil or similar dielectric liquid and a heat exchanger (not shown), where the copper pipe **1105** is inserted inside the hollow inner electrode **1110** for cooling the inner electrode **1110** and removing heat from the carbon source **1130**. The device **1100** also includes a top spring **1115** to push the top quartz lid **1125** and compress the carbon source **1130** in space **1131** and a bottom spring **1140** to push the bottom quartz lid **1135** and compress the carbon source **1130** in space **1131**. The carbon source **1130** is moved into the space for joule heating the carbon source **1131** to undergo conversion to graphene. The carbon source **1130** may be inserted and removed in a batch mode, where carbon source **1130** is inserted in the radial space **1131**, is joule heated to convert to graphene, and removed from the radial space **1131**. The carbon source **1130** may also be inserted and removed with an extrusion screw that replaces the confinement provided by the quartz lids.

(55) The electrical current flows from the inner **1110** to outer electrode **1120** or vice versa in a radial manner, the carbon source **1130** is confined and is joule heated by the outer graphite electrodes **1110** and **1120**. In some embodiments, the inner **1110** and outer electrodes **1120** may be liquid cooled.

(56) Radiation cooling may occur from the carbon source **1130** to the outer graphite electrode **1120** with high efficiency as the graphite is a black body heat receptor. The outer graphite electrode **1120**

is also heated by conduction from the carbon source **1130** but may then re-radiate the heat from its outer surface and may be cooled additionally by convection or conduction from the outside. The outer electrode **1120** may be porous graphite, which allows for gas and heat to leave the carbon source more efficiently. The outer electrode **1120** may also have holes spread along its surface to allow gas escape from the inside to the outside of the electrode. No consumable quartz tubes are needed as the graphite electrode confines the carbon source and converted graphene. The graphite electrodes **1120**, **1110** may be used almost indefinitely with minimal wear.

(57) Referring to FIG. **11B**, illustrated therein is a diagram of a cross section view of a device **1101** for continuous synthesis of graphene utilizing a radial electrode arrangement, and an extrusion screw **1145** which moves the carbon source **1150** in and out of the radial space of the device **1101**, according to an embodiment. The device **1101** includes an inner **1155** and an outer **1160** electrode aligned in a radial manner. The device **1101** also includes an extrusion screw **1145** that moves the carbon source powder **1150** from an external reservoir (not shown) into a ceramic tube conduit **1170** and then into the radial space between the radial electrodes **1155**, **1160**. The tube **1170** is the container and the carbon source **1150** is moved into the entry opening **1171** of the container. The extrusion screw **1145** has a hollow shaft **1180**. The inner electrode **1155** may be fixed to the shaft of the extrusion screw **1145** and an electrical connection may be passed through the hollow shaft. The positive terminal **1172** and negative terminal **1173** are connected to a power supply for operation in either alternating current (AC), direct current (DC), or a combination of AC and DC. During operation, the space **1175** for joule heating the carbon source **1150** between the electrodes **1155**, **1160** is filled with the carbon source **1150** and the joule heating of the carbon source **1150** is executed. The synthesized graphene **1165** in the space for joule heating the carbon source **1150** between the electrodes **1155**, **1160** is removed through an exit opening. The movement of the carbon source **1150** is continuous and is not obstructed by the electrodes **1155**, **1160**. Further, the flow of carbon source **1150** is lengthwise down the tube, whereas the electrical current flow is in the radial direction.

(58) Referring to FIG. **12**, illustrated therein is a flow chart of a method for synthesis of graphene, according to an embodiment. The method **1200** includes moving a carbon source in a first direction into a space for joule heating the carbon source, at **1205**. The method **1200** optionally includes compressing the carbon source, at **1206**. The method **1200** also includes applying an electrical current to the carbon source in a second direction to convert at least part of the carbon source into graphene with the at least two electrodes positioned to allow movement of the carbon source into the space for joule heating the carbon source, at **1210**. The method **1200** at **1210** may optionally include an additional step at **1215** which includes applying a first electrical current to the carbon source at a lower voltage for removing moisture and volatile materials from the carbon source. The lower voltage generates lower process temperatures that remove moisture and volatiles from the carbon source but does not convert the carbon source into graphene. The method **1200** at **1210** may optionally include an additional step at **1220** which includes applying a second electrical current at a higher voltage for converting at least part of the carbon source to graphene. The higher voltage enables joule heating of the carbon source and conversion into graphene. The method **1200** may optionally include removing unconverted carbon from the graphene, at **1230**.

(59) While the above description provides examples of one or more apparatus, methods, or systems, it will be appreciated that other apparatus, methods, or systems may be within the scope of the claims as interpreted by one of skill in the art.

Claims

1. A device for synthesis of graphene, the device comprising: a container having a space for holding a carbon source, wherein the container has an entry opening for receiving the carbon source; at least two electrodes for applying an electrical current through the space for joule heating the carbon

source, wherein the space for joule heating the carbon source is between the at least two electrodes, wherein the at least two electrodes are positioned to radially apply an electrical current through the space for joule heating the carbon source; and a movement component for moving the carbon source, with respect to the container, into the entry opening in a first direction and the at least two electrodes apply the electrical current in a second direction, wherein the first direction is not the same as the second direction.

2. The device of claim 1, further comprising an exit opening positioned relative to the at least two electrodes to allow for moving the graphene out of the space for joule heating the carbon source, and wherein the entry opening is positioned relative to the at least two electrodes to allow for moving of the carbon source into the space for joule heating the carbon source while applying the electrical current.

3. The device of claim 1, further comprising a power supply connected to the electrodes for passing the electrical current through the electrodes to convert at least part of the carbon source in the space for joule heating the carbon source into graphene.

4. The device of claim 1, wherein the at least two electrodes include vents for escape of gas when the electrical current is applied to the carbon source.

5. The device of claim 1, wherein the space for joule heating the carbon source is surrounded by at least one quartz wall.

6. The device of claim 1 further comprising a carbon source reservoir for holding the carbon source prior to moving into the space for joule heating the carbon source and a graphene reservoir for collecting the graphene after moving out of the space for joule heating the carbon source.

7. The device of claim 1 further comprising a compression component for compressing the carbon source.

8. The device of claim 7 wherein the compression component is a compression corkscrew.

9. The device of claim 1, wherein the at least two electrodes are positioned opposite to each other.

10. The device of claim 1, wherein the graphene is turbostratic graphene.

11. A device for synthesis of graphene, the device comprising: an inner electrode and an outer electrode; a space for joule heating the carbon source for a carbon source between the inner electrode and the outer electrode, wherein the inner electrode and outer electrode are positioned to radially apply an electrical current through the space for joule heating the carbon source; an entry opening positioned relative to the inner electrode and outer electrode to allow for moving of the carbon source into the space for joule heating the carbon source; and a power supply connected to the inner electrode and outer electrode for passing the electrical current through the inner electrode and outer electrode to convert at least part of the carbon source in the space for joule heating the carbon source into graphene.

12. The device of claim 11, wherein the outer electrode surrounds the space for joule heating the carbon source.

13. The device of claim 11 further comprising a cooling component for cooling at least one of the group comprising the inner electrode and the outer electrode.

14. A device for synthesis of graphene, the device comprising: a container having a space for holding a carbon source, wherein the container has an entry opening for receiving the carbon source; at least two ring electrodes for applying an electrical current through the space for joule heating the carbon source, wherein the space for joule heating the carbon source is between the at least two ring electrodes; and a movement component for moving the carbon source, with respect to the container, into the entry opening in the same direction the at least two ring electrodes apply the electrical current.

15. The device of claim 14 further comprising an exit opening positioned relative to the at least two ring electrodes to allow for moving the graphene out of the space for joule heating the carbon source, and wherein the entry opening is positioned relative to the at least two electrodes to allow for moving of the carbon source into the space for joule heating the carbon source while applying

the electrical current.

16. The device of claim 14 further comprising a power supply connected to the electrodes for passing the electrical current through the at least two ring electrodes to convert at least part of the carbon source in the space for joule heating the carbon source into graphene.

17. The device of claim 14 further comprising pre-treatment ring electrodes for heating the carbon source to temperatures between 400° C.-800° C.

18. The device of claim 14, wherein the at least two electrodes are configured to apply an electric current which heats the carbon source to temperatures between 2800° C.-3200° C.

19. The device of claim 14 further comprising a carbon source reservoir for holding the carbon source prior to moving into the space for joule heating the carbon source and a graphene reservoir for collecting the graphene after moving out of the space for joule heating the carbon source.
